

STUDENT'S NAME: _____

TEACHER'S NAME: _____

BAULKHAM HILLS HIGH SCHOOL

.

**TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION**

2007

MATHEMATICS

EXTENSION 2

GENERAL INSTRUCTIONS:

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using blue or black pen.
- Start each question on a new page.
- Write your Student Number at the top of each page.
- Calculators may be used.
- A table of standard integrals is provided.
- **ALL** necessary working should be shown in every Question.

QUESTION 1 (15 marks)

Marks

- (a) Simplify $\sin(A - B) + \sin(A + B)$ and hence find

$$\int \sin 5x \cos 3x \, dx \quad 3$$

- (b) Find real constants A, B, C such that:

$$\frac{x^2 + 5x + 2}{(x^2 + 1)(x + 1)} \equiv \frac{Ax + B}{x^2 + 1} + \frac{C}{x + 1}$$

and hence find $\int \frac{x^2 + 5x + 2}{(x^2 + 1)(x + 1)} dx \quad 3$

- (c) Find $\int \sin^{-1} x \, dx \quad 3$

- (d) Find $\int \sqrt{\frac{1+x}{1-x}} dx \quad 3$

- (e) Evaluate $\int_0^1 x^5 e^{x^3} dx \quad 3$

QUESTION 2 (15 marks)

(a) If $z = 2\sqrt{3}i - 2$ find:

(i) $|z|$ 1

(ii) $\arg z$ 1

(iii) $\operatorname{Re}(1+2i)\bar{z}$ 2

b) If ω is a complex root of the equation

$$z^3 = 1:$$

(i) Show that $1 + \omega + \omega^2 = 0$ 2

(ii) Find the value of $(1 - \omega)(1 - \omega^2)(1 - \omega^4)(1 - \omega^8)$ 3

(c) Find the equation of the locus of z where $|z - 2i| = \operatorname{Im} z$. 3

(d) Given $z = 2 - i$, find real values of a and b such that: 3

$$az + \frac{b}{z} = 1$$

QUESTION 3 (15 marks)

Marks

(a) Sketch the curve:

$$f(x) = \frac{(x+1)(x+3)}{x} \text{ showing the stationary points and asymptotes.}$$

3

Hence draw neat half page sketches of:

(i) $y = \frac{1}{f(x)}$

2

(ii) $y = |f(x)|$

1

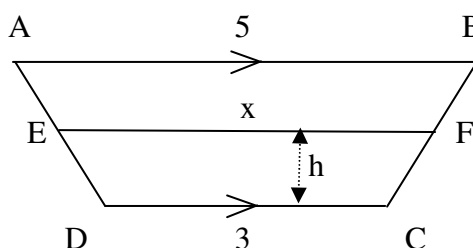
(iii) $y = \sqrt{f(x)}$

1

(iv) $y = \log_e f(x)$

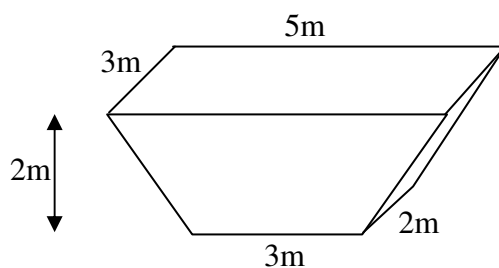
2

(b)



A trapezium ABCD has parallel sides $AB = 5$ m and $CD = 3$ m, 2 m apart. E lies on AD and F lies on BC such that EF is parallel to DC. The distance from EF to DC is h m and $EF = x$ show that:
 $x = 3 + h$.

1

NOT TO
SCALE

The diagram is of a waste bin with a rectangular base of side 3m and 2m. Its top is also rectangular, parallel to the base with dimensions 5m and 3m. The bin has a depth of 2m, each of its four sides are trapeziums. Find the volume of the bin.

5

QUESTION 4 (15 marks)

Marks

- (a) (i) Prove that the normal at point $P\left(cp, \frac{c}{p}\right)$ on the curve $xy = c^2$

$$\text{is } p^3x - py = c(p^4 - 1) \quad 2$$

- (ii) The normal at P meets the hyperbola again at point $Q\left(cq, \frac{c}{q}\right)$. Prove that $p^3q = -1$ 2

- (iii) The tangent at P meets the y axis at R. 3

Show that the area of the triangle PQR is:

$$A = \frac{c^2}{2} \left(p^2 + \frac{1}{p^2} \right)^2$$

and hence find the minimum area of this triangle. 2

- (b) (i) Evaluate $\int_0^{\frac{\pi}{2}} (\sin t)^{2k} \cos t \, dt$ 1

- (ii) Noting that $(\cos t)^{2n+1} = \cos t (1 - \sin^2 t)^n$ and by using the Binomial Theorem to expand $(1 - \sin^2 t)^n$ where n is a positive integer, show that

$$\int_0^{\frac{\pi}{2}} (\cos t)^{2n+1} dt = \sum_{r=0}^n (-1)^r \frac{1}{2r+1} \cdot {}^nC_r \quad 3$$

- (iii) Use the result of part (ii) to evaluate

$$\int_0^{\frac{\pi}{2}} \cos^7 t \, dt \quad 2$$

QUESTION 5 (15 marks)

Marks

- (a) The quadratic equation $x^2 - x + k = 0$ where k is a real number has 2 distinct positive roots α and β .

Show that:

(i) $0 < k < \frac{1}{4}$ 2

(ii) $\alpha^2 + \beta^2 = 1 - 2k$ and hence deduce that $\alpha^2 + \beta^2 > \frac{1}{2}$ 3

(iii) $\frac{1}{\alpha^2} + \frac{1}{\beta^2} > 8$ 2

- (b) Let α , β and γ be the roots of:

$$x^3 - 7x^2 + 18x - 7 = 0$$

Find the polynomial with roots:

(i) $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ 2

(ii) $(1 + \alpha^2), (1 + \beta^2), (1 + \gamma^2)$ 3

- (c) By considering the stationary values of:

$$f(x) = x^3 - 3px^2 + 4q, \text{ where } p \text{ and } q \text{ are positive real constants, show}$$

that the equation $f(x) = 0$ has 3 real distinct roots if

$$p^3 > q. \quad \text{3}$$

QUESTION 6 (15 marks)

Marks

- (a) A particle of mass m is moving vertically in a resisting medium in which the resistance to the motion has a magnitude of $\frac{1}{10} m v^2$ where the particle has speed $u \text{ ms}^{-1}$. The acceleration due to gravity is $g \text{ ms}^{-2}$.

- (i) If the particle falls vertically downwards from rest, show that its acceleration is given by:

$$a = g - \frac{1}{10} v^2.$$

Hence show that its terminal speed $V \text{ ms}^{-1}$ is given by

$$V = \sqrt{10g}. \quad 2$$

- (ii) If the particle is projected vertically upwards with speed $V \tan \alpha \text{ ms}^{-1}$ ($0 < \alpha < \frac{\pi}{2}$) show that its acceleration $a \text{ ms}^{-2}$ is given by

$$a = -\left(g + \frac{v^2}{10}\right)$$

Hence show that it reaches a maximum height H metres given by :

$$H = 5 \log_e \sec^2 \alpha \quad 4$$

and that it returns to its point of projection with speed

$$V \sin \alpha \text{ ms}^{-1}. \quad 4$$

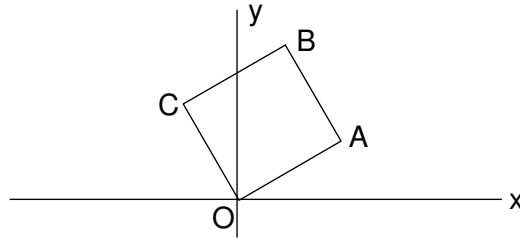
- (b) The roots of the equation $4x^3 - 36x^2 + 107x + k = 0$ are in arithmetic progression, find:

- (i) k 2
- (ii) the roots of the equation. 3

QUESTION 7 (15 marks)

Marks

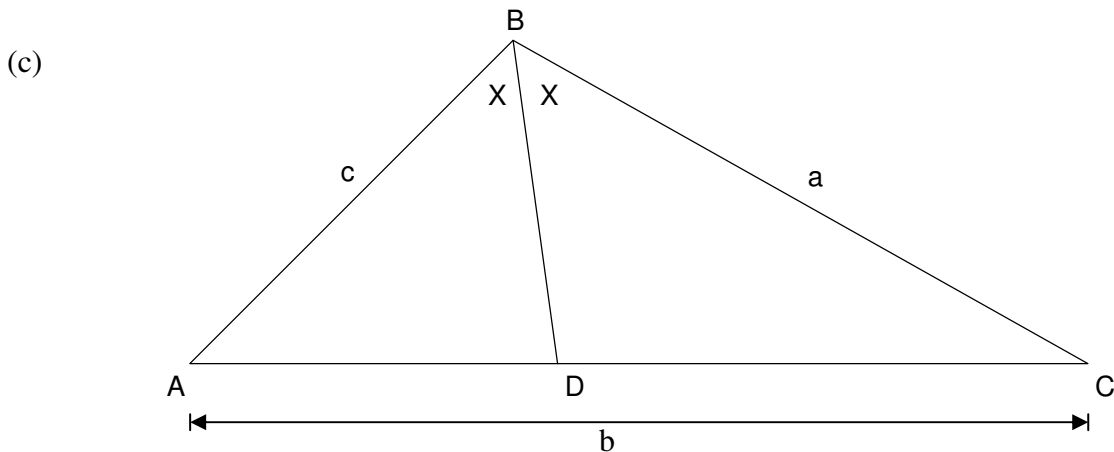
- (a) In the square OABC shown below, the point A represents $4 + 3i$. What complex numbers do the points B and C represent.



3

- (b) (i) Determine the real values of k for which the equation:

$$\frac{x^2}{19-k} + \frac{y^2}{7-k} = 1$$
defines an ellipse and a hyperbola respectively. 2
- (ii) Sketch the curve corresponding to the value of $k = 3$, showing foci, directrices and where the curve cuts the coordinate axes. 4
- (iii) Describe how the shape of this curve changes as k varies from 3 to 7. 1



In $\triangle ABC$, BD bisects $\angle ABC$ as shown in the diagram.

- (i) By considering the area of $\triangle ABC$, show that 2

$$BD = \frac{2ac \cos x}{a+c}$$

- (ii) Show that : $\cos x = \frac{1}{2} \sqrt{\frac{(a+c)^2 - b^2}{ac}}$ 2

- (iii) Hence show that $BD = \frac{\sqrt{ac}}{a+c} \sqrt{(a+c)^2 - b^2}$ 1

QUESTION 8 (15 marks)

Marks

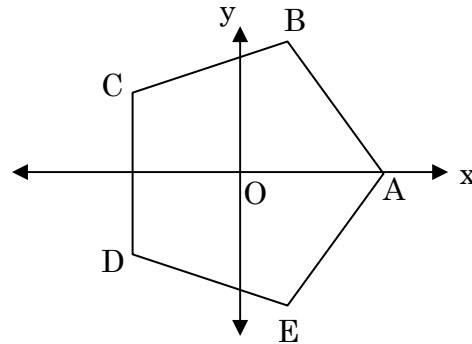
- (a) If $I_n = \int_1^e x^3 (\log_e x)^n dx$ for $n = 0, 1, 2, \dots$ show that

$$I_n = \frac{e^4}{4} - \frac{n}{4} I_{n-1} \quad \text{and hence find the value of}$$

$$\int_1^e x^3 (\log_e x)^2 dx$$

4

(b)



In the diagram, the complex numbers z_0, z_1, z_2, z_3 and z_4 are represented by the vertices of a regular pentagon with center O and vertices A, B, C, D and E respectively. Given that $z_0 = 2$

- (i) Express z_2 in modulus argument form 1
- (ii) Find the value of $(z_2)^5$ 2
- (iii) Show that the perimeter of the pentagon is $20 \sin \frac{\pi}{5}$ 2
- (c) (i) Use De Moivre's Theorem to find expressions for $\cos 5\theta$ and $\sin 5\theta$ and hence show that: 3

$$\tan 5\theta = \frac{5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta}{1 - 10 \tan^2 \theta + 5 \tan^4 \theta}$$

- (ii) By considering the equation: 3

$$x^5 - 5x^4 - 10x^3 + 10x^2 + 5x - 1 = 0$$

$$\text{Prove that } \tan \frac{\pi}{20} + \tan \frac{9\pi}{20} + \tan \frac{17\pi}{20} + \tan \frac{33\pi}{20} = 4$$

EXT 2 TRIAL 2007

Da)

$$\sin(A-B) + \sin(A+B) = \sin A \cos B - \cos A \sin B + \sin A \cos B + \cos A \sin B = 2 \sin A \cos B$$

Let $A = 5x$ $B = 3x$

$$\therefore \int \sin 5x \cos 3x = \frac{1}{2} \int \sin 2x dx + \frac{1}{2} \int \sin 8x dx$$

$$= -\frac{1}{4} \cos 2x - \frac{1}{16} \cos 8x + C$$

b) $(Ax+B)(x+1) + C(x^2+1) \equiv x^2 + 5x + 2$

$x = -1$ $2C = -2 \therefore C = -1$

Coef of x^2 $A+C = 1 \therefore A = 2$

Constant term $B+C = 2 \therefore B = 3$

$$\therefore I = \int \frac{2x+3}{x^2+1} dx - \int \frac{dx}{x+1}$$

$$= \ln(x^2+1) + 3 \ln|x| - \ln|x+1| + C$$

c) $I = \int \sin^{-1} x dx$

Let $u = \sin^{-1} x$ $u' = 1$

$u' = \frac{1}{\sqrt{1-x^2}}$ $u = x$

$$\therefore I = x \sin^{-1} x - \int \frac{x}{\sqrt{1-x^2}} dx$$

$$= x \sin^{-1} x + \sqrt{1-x^2} + C$$

d) $\int \sqrt{\frac{1+x}{1-x}} dx$

$$= \int \frac{1+x}{\sqrt{1-x^2}} dx$$

$$= \int \frac{dx}{\sqrt{1-x^2}} + \int \frac{x dx}{\sqrt{1-x^2}}$$

$$= \sin^{-1} x - \sqrt{1-x^2} + C$$

$$\int_0^1 x^5 e^{x^3} dx$$

Let $u = \frac{x^3}{3}$ $u' = x^2 e^{x^3}$

$u' = x^2$ $v = e^{x^3}$

$$\therefore I = \left[\frac{x^3}{3} e^{x^3} \right]_0^1 - \int_0^1 x^2 e^{x^3} dx$$

$$= \frac{e}{3} - 0 - \left[\frac{e}{3} \right]_0^1$$

$$= \frac{e}{3} - \frac{e}{3} + \frac{1}{3}$$

$$= \frac{1}{3}$$

$$\textcircled{2} a) i) |z| = \sqrt{(2\sqrt{3})^2 + (-2)^2} \\ = \sqrt{12+4} = 4$$

$$ii) \arg z = \sqrt{3} \quad \cos \theta = -\frac{1}{2} \\ \therefore \arg z = \frac{2\pi}{3}$$

$$iii) \operatorname{Re} (1+2i)(-2\sqrt{3}i-2) \\ = -2+4\sqrt{3}$$

$$b) ii) z^3 = 1$$

$$z^3 - 1 = 0$$

$$(z-1)(z^2+z+1) = 0$$

if ω is a complex root $\omega \neq 1$

$$\therefore \omega^2 + \omega + 1 = 0$$

$$ii) (1-\omega)(1-\omega^2)(1-\omega^4)(1-\omega^8)$$

$$\text{now } \omega^4 = \omega^2 \cdot \omega = \omega$$

$$\omega^8 = (\omega^2)^2 \cdot \omega^2 = \omega^2$$

$$\therefore = (1-\omega)(1-\omega^2)(1-\omega)(1-\omega^2)$$

$$= (1-\omega-\omega^2+\omega^3)^2$$

$$\text{now } \omega + \omega^2 = -1 \quad \omega^3 = 1$$

$$\therefore = (1+1)^2 = 4$$

$$c) |z-2i| = 9 \Rightarrow z$$

$$\text{Let } z = x+iy$$

$$\therefore \sqrt{x^2 + (y-2)^2} = 3$$

$$\therefore x^2 + y^2 - 4y + 4 = 9$$

$$\therefore x^2 - 4y + 4 = 0$$

$$d) z = 2-i$$

$$az + \frac{b}{z} = 1$$

$$\therefore a(2-i) + \frac{b}{2-i} = 1$$

$$\times (2-i) \text{ by } (2-i)$$

$$a(2-i)^2 + b = 2-i$$

$$a(3-4i) + b = 2-i$$

$$\therefore (3a+b-2) + i(-4a+1) = 0$$

$$\therefore a = \frac{1}{4} \quad *$$

$$\frac{3}{4} + b - 2 = 0$$

$$\therefore b = \frac{5}{4} \quad *$$

$$\text{OR } a(2-i) + \frac{b}{2-i} = 1$$

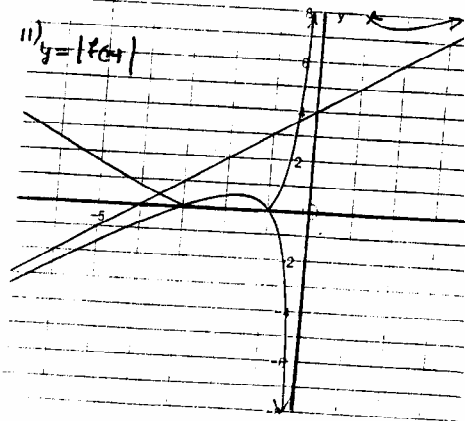
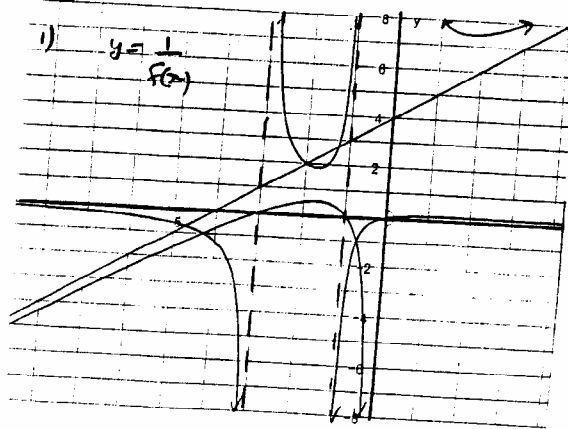
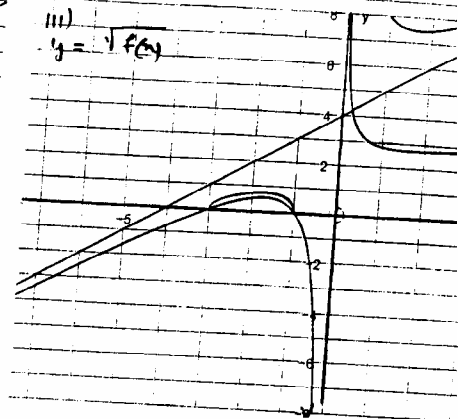
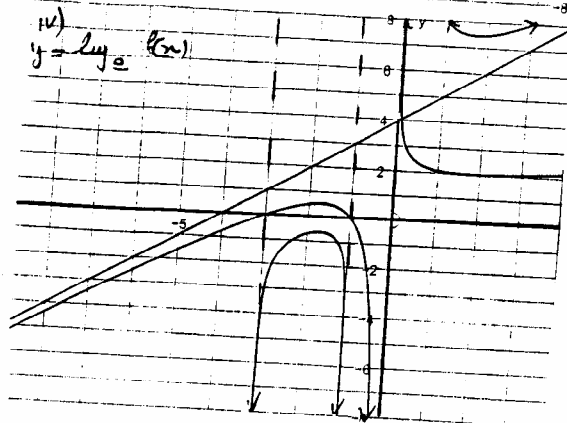
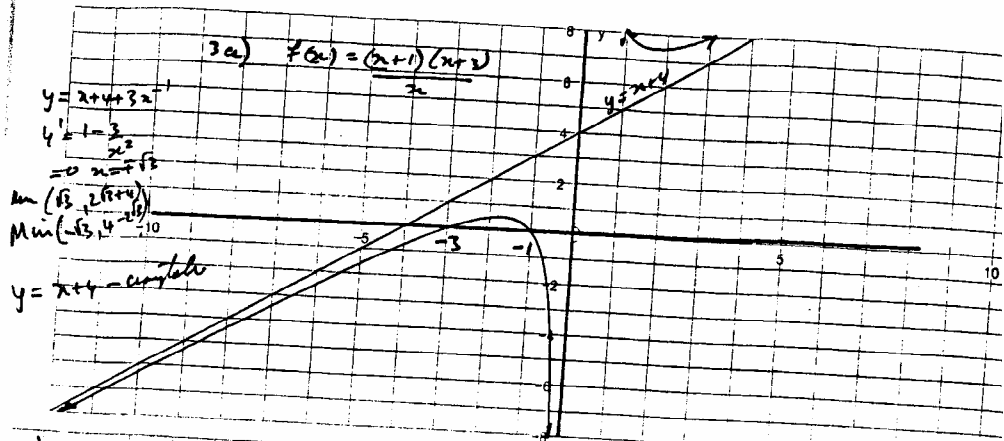
$$\frac{2a-ai}{2-i} + \frac{2b+ib}{2-i} = 1$$

$$(10a+2b-5) + i(b-5a) = 0$$

$$10a+2b=5$$

$$b-5a=0$$

$$\therefore b = \frac{5}{4} \quad a = \frac{1}{4}$$



36

$$\text{Let } x = a + bh$$

$$\text{when } h=0, x=3, \quad h=2, x=5$$

$$\therefore a=3$$

$$x = 3 + bh$$

$$5 = 3 + 2b$$

$$\therefore b=1$$

$$\therefore x = 3 + h$$

$$\text{Similarly } y = 2 + \frac{h}{2}$$

$$\therefore SA = (3+h)(2+\frac{h}{2})$$

$$SV = (3+h)(2+\frac{h}{2}) dh$$

$$\therefore V = \int_0^2 (3+h)(2+\frac{h}{2}) dh$$

$$= \frac{1}{2} \int_0^2 (12 + 7h + h^2) dh$$

$$= \frac{1}{2} \left[12h + \frac{7h^2}{2} + \frac{h^3}{3} \right]_0^2$$

$$= \frac{1}{2} \left[24 + 14 + \frac{8}{3} \right] - 0$$

$$= 20\frac{1}{3} \text{ units.}$$

4a) i) $y = c^2 x^{-1}$

$$y' = \frac{-c^2}{x^2} = -\frac{1}{p^2}$$

$\therefore$ grad of normal is p^2

$$\therefore n \Rightarrow y - \frac{c}{p} = p^2(x - cp)$$

$$\therefore p^3 x - py = c(p^4 - 1)$$

ii) Subst $x = cp$ $y = \frac{c}{p} \Rightarrow n$

$$\therefore cp^3 q - \frac{pc}{q} = cp^4 - c$$

$$\therefore p^3 q^2 - p = p^4 q - q$$

$$p^3 q^2 - p^4 q = p - q$$

$$p^3 q(q - p) = (p - q)$$

$$\therefore p^3 q = -1$$

iii) equation of tangent is

$$x + p^2 y = 2cp$$

subst $x=0 \therefore y = \frac{2c}{p}$

$$\therefore R(0, \frac{2c}{p})$$

$$P(cp, \frac{c}{p})$$

$$Q(-\frac{c}{p^3}, -cp^3)$$

Area of $\Delta PQR = \frac{1}{2} PR \times QR$

$$= \frac{1}{2} \sqrt{(cp)^2 + (\frac{c}{p})^2} \cdot \sqrt{(cp + \frac{c}{p^3})^2 + (\frac{c}{p} + cp^3)^2}$$

$$= \frac{c^2}{2} \sqrt{p^2 + \frac{1}{p^2}} \cdot \sqrt{(p + \frac{1}{p^3})^2 + (\frac{1}{p} + p^3)^2}$$

$$= \frac{c^2}{2} \sqrt{\frac{(p^4+1)}{p^2}} \cdot \sqrt{\left(\frac{p^4+1}{p^3}\right)^2 + \left(\frac{p^4+1}{p}\right)^2}$$

$$= \frac{c^2}{2} \sqrt{\frac{(p^4+1)}{p^2}} \cdot \sqrt{(p^4+1)^2 \left(\frac{1}{p^6} + \frac{1}{p^2}\right)}$$

$$= \frac{c^2}{2} \frac{\sqrt{p^4+1}}{p} \cdot (p^4+1) \sqrt{\frac{p^4+1}{p^6}}$$

$$= \frac{c^2}{2} \frac{(p^4+1)^2}{p^4}$$

$$= \frac{c^2}{2} \left(\frac{p^4+1}{p^2}\right)^2$$

$$A = \frac{c^2}{2} \left(p^2 + \frac{1}{p^2}\right)^2$$

$$\frac{dA}{dp^2} = \frac{c^2}{2} 2\left(p^2 + \frac{1}{p^2}\right) \left(1 - \frac{1}{p^4}\right)$$

$$= 0 \text{ when } p^2 = 1$$

$$\therefore A = \frac{c^2}{2} \cdot 2^2 = 2c^2$$

b) $\int_0^{\frac{\pi}{2}} (\cos t)^{2k+1} \cos t dt$

$$= \left[\frac{(\cos t)^{2k+2}}{2k+2} \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{2k+2}$$

$$\int_0^{\frac{\pi}{2}} (\cos t)^{2n+1} dt = \int_0^{\frac{\pi}{2}} \cos t (1 - \sin^2 t)^n dt$$

$$= \int_0^1 \cos t - \sin^2 t \cos t + \dots + (-1)^n \sin^{2n} t \cos t dt$$

$$= \frac{1}{2n+1} - \frac{1}{2n+1} \frac{1}{2n+1} + \frac{1}{2n+1} \frac{1}{2n+1} \dots$$

$$= \sum_{r=0}^n (-1)^r \frac{1}{2r+1} \cdot \frac{1}{2r+1}$$

$$\int_0^{\frac{\pi}{2}} \cos^7 t dt \rightarrow n=3 \text{ (not?)}$$

$$\therefore I = 1 - \frac{3C_1}{3} + \frac{3C_2}{5} - \frac{3C_3}{7}$$

Q5

(a) $x^2 - x + k = 0$
 $\alpha + \beta = 1, \alpha\beta = k$
 $\alpha, \beta > 0 \therefore k > 0$ ✓
 α, β are real, distinct $\therefore \Delta > 0$
 $1 - 4 \cdot 1 \cdot k > 0$
 $\frac{1}{4} > k$ ✓

(i) $\therefore 0 < k < \frac{1}{4}$

(ii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
 $= 1 - 2k$ ✓

Since $0 < k < \frac{1}{4}$
 $0 > -2k > -\frac{1}{2}$ ✓
 $1 > 1 - 2k > 1 - \frac{1}{2} = \frac{1}{2}$ ✓

$\therefore \alpha^2 + \beta^2 = 1 - 2k > \frac{1}{2}$

(iii) $k < \frac{1}{4}, \frac{1}{k} > 4$
 $\frac{1}{k^2} > 4^2$, ie $\frac{1}{\alpha^2\beta^2} > 16$ ✓

$\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\alpha^2 + \beta^2}{\alpha^2\beta^2} > \frac{1}{2} \times 16$
 > 8 ✓

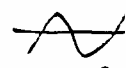
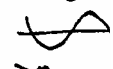
(b) (i) Let $y = \frac{1}{x}, x = \frac{1}{y}$
 $\therefore$ Eq is $(\frac{1}{y})^3 - 7(\frac{1}{y})^2 + 18(\frac{1}{y}) - 7 = 0$
 $1 - 7y + 18y^2 - 7y^3 = 0$
 $\text{or } 7x^3 - 18x^2 + 7x - 1 = 0$ ✓✓

(ii) Let $y = 1 + x^2$
 $x^2 = y - 1$ ✓
 $\sqrt{y-1}(y-1) - 7(y-1) + 18\sqrt{y-1} - 7 = 0$
 $\sqrt{y-1}(y+17) = 7y$
 $(y-1)(y^2 + 34y + 289) = 49y^2$
 $y^3 + 34y^2 + 289y - y^2 - 34y - 289 = 49y^2$

or $x^3 - 16x^2 + 255x - 289 = 0$

(c) $\varphi(x) = x^3 - 3px^2 + 4q$
 $\varphi'(x) = 3x^2 - 6px$
S.P when $\varphi'(x) = 0$ $3x(x-2p) = 0$
 $x = 0, 2p$

$\varphi(0) = 4q > 0$ since $q > 0$
 $\varphi(2p) = (2p)^3 - 3p(2p)^2 + 4q$
 $= 4q - 4p^3$

For 3 real distinct roots 
product of y values < 0 
 $\therefore 4q - 4p^3 < 0$ since $4q > 0$

Q6

a) Downward $\downarrow +$
 $\uparrow \frac{1}{10}mv^2$
 $\downarrow mg$
 $mg - \frac{1}{10}mv^2 = ma$
 $\therefore a = g - \frac{1}{10}v^2$

terminal velocity when $a = 0$
 $\therefore V^2 = 10g$
 $V = \sqrt{10g}$

(ii) Upwards $\uparrow +$
 $\downarrow \frac{1}{10}mv^2$
 $\downarrow mg$
 $-(mg + \frac{1}{10}mv^2) = ma$
 $\therefore a = -(g + \frac{v^2}{10})$
 $v \frac{dv}{dx} = -(g + \frac{v^2}{10})$
 $-\int \frac{10v}{10g + v^2} dv = \int dx$
 $\text{v to } \infty \quad x = -5 \ln(10g + v^2) + C$
 $\therefore H = [5 \ln(10g + v^2)]_0^{\infty}$
 $= 5 \ln(10g + v^2 + \infty) - 5 \ln$
 $= 5 \ln \frac{10g + 10g + \infty}{10g}$

$$H = 5 \ln(1 + \tan^2 \alpha) = 5 \ln \sec^2 \alpha$$

For downward, let speed at pt of projection be u

$$u \frac{dv}{dx} = g - \frac{1}{10} v^2$$

$$\int_0^H \frac{10v}{10g - v^2} dv = \int_0^H dx$$

$$[-5 \ln(10g - v^2)]_0^H = H$$

$$5 \ln 10g - 5 \ln(10g - u^2) = H$$

$$5 \ln \sec^2 \alpha = 5 \ln \frac{10g}{10g - u^2}$$

$$10g - u^2 = \frac{10g}{\sec^2 \alpha} = 10g \cos^2 \alpha$$

$$\begin{aligned} u^2 &= 10g(1 - \cos^2 \alpha) \\ &= 10g \sin^2 \alpha \\ &= v^2 \sin^2 \alpha \end{aligned}$$

$$u = v \sin \alpha$$

$$(b) \quad 4x^3 - 36x^2 + 107x + k = 0$$

Let roots be $a-d, a, a+d$

$$\text{Sum} = 3a = \frac{36}{4} \therefore a = 3 \checkmark$$

3 is a root

$$\therefore 4(3)^3 - 36(3)^2 + 107(3) + k = 0$$

$$k = -105 \checkmark$$

$$\begin{array}{r} 4x^2 - 24x + 35 \\ x-3 \overline{) 4x^3 - 36x^2 + 107x - 105} \\ \underline{4x^3 - 12x^2} \\ -24x^2 + 107x \\ \underline{-24x^2 + 72x} \\ 35x - 105 \\ \underline{35x - 105} \\ 0 \end{array}$$

$$\therefore (x-3)(4x^2 - 24x + 35) = (x-3)(2x-7)(2x-5)$$

$$x = \frac{5}{2}, 3, \frac{7}{2} \checkmark \checkmark$$

(a)

or

$$x = -5 \ln(10g - v^2) + C$$

$$C = 5 \ln(10g) \checkmark$$

$$x = 5 \ln \frac{10g}{10g - v^2} \checkmark$$

$$5 \ln \sec^2 \alpha = 5 \ln \frac{10g}{10g - v^2}$$

$$\sec^2 \alpha = \frac{10g}{10g - v^2}$$

$$10g - v^2 = 10g \cos^2 \alpha \checkmark$$

$$\begin{aligned} v^2 &= 10g(1 - \cos^2 \alpha) \\ &= v^2 \sin^2 \alpha \checkmark \end{aligned}$$

$$3a = \frac{36}{4} \quad a = 3 \checkmark$$

$$a^2 - ad + a^2 + ad + a^2 - d^2 = \frac{107}{4}$$

$$3a^2 - d^2 = \frac{107}{4}$$

$$d^2 = 27 - \frac{107}{4} = \frac{1}{4}$$

$$d = \pm \frac{1}{2} \checkmark$$

$\therefore$ Roots are $3 - \frac{1}{2}, 3, 3 + \frac{1}{2}$

$$2\frac{1}{2}, 3, 3\frac{1}{2} \checkmark$$

$$-\frac{k}{4} = \frac{5}{2} \times 3 \times \frac{7}{2} \quad k = -105 \checkmark$$

1) C rep by $i(4+3i)$
 $\therefore$ C represents $-3+4i$

Br rep by $(-3+4i)(4+3i)$
 $= 1+7i$

b) i) for ellipse

$19-k > 0$ and $7-k > 0$
 $k < 19$ $k < 7$

$\therefore k < 7$ represents an ellipse

for hyp $19-k > 0$ and $7-k < 0$
 $k < 19$ $k > 7$

$\therefore 7 < k < 19$

or $19-k < 0$ $7-k > 0$
 $k > 19$ $k < 7$

no such

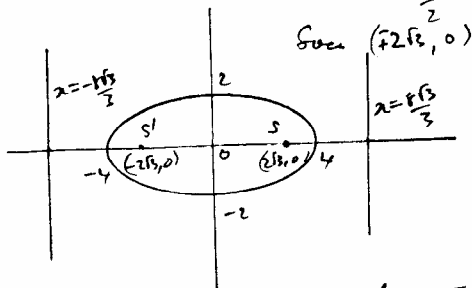
$7 < k < 19$ represents a hyperbola

ii) $k=3 \Rightarrow$

$\frac{x^2}{16} + \frac{y^2}{4} = 1$

$a=4$ $b=2$ $b^2 = a^2(1-e^2)$
 $4 = 16(1-e^2)$

$e = \frac{\sqrt{3}}{2}$
 Focus $(\pm 2\sqrt{3}, 0)$



and $n = \frac{8}{\sqrt{3}}$
 $n = \frac{8\sqrt{3}}{3}$

c) i) $AREA = \frac{1}{2} \cdot a \cdot c \sin 2x$

also $AREA = \frac{1}{2} c \cdot BD \sin x + \frac{1}{2} a \cdot BD \sin$

$= \frac{BD \sin x (a+c)}{2}$

$\therefore BD \sin x (a+c) = \frac{1}{2} ac \cdot 2 \sin x \cos x$

$\therefore BD = \frac{2ac \cos x}{a+c}$

ii) $\cos 2x = \frac{a^2+c^2-b^2}{2ac}$

$\therefore 2\cos^2 x - 1 = \frac{a^2+c^2-b^2}{2ac}$

$\cos^2 x = \frac{1}{2} \left(\frac{a^2+c^2-b^2}{2ac} + 1 \right)$
 $= \frac{a^2+c^2-b^2+2ac}{4ac}$

$= \frac{(a+c)^2 - b^2}{4ac}$

$= \frac{(a+c)^2 - b^2}{4ac}$

$\therefore \cos x = \frac{1}{2} \sqrt{\frac{(a+c)^2 - b^2}{ac}}$

iii) $BD = \frac{2ac \cos x}{a+c}$

$= \frac{2ac}{a+c} \cdot \frac{1}{2} \sqrt{\frac{(a+c)^2 - b^2}{ac}}$

$= \frac{\sqrt{ac} \sqrt{(a+c)^2 - b^2}}{a+c}$

$$8a) I_n = \int_1^e x^3 (\ln x)^n dx$$

$$\text{Let } u = (\ln x)^n \quad v' = x^3$$

$$u' = n(\ln x)^{n-1} \cdot \frac{1}{x} \quad v = \frac{x^4}{4}$$

$$\therefore I_n = \left[\frac{x^4}{4} (\ln x)^n \right]_1^e - \frac{n}{4} \int_1^e x^3 (\ln x)^{n-1} dx$$

$$= \frac{e^4}{4} - 0 - \frac{n}{4} I_{n-1}$$

$$\therefore I_n = \frac{e^4}{4} - \frac{n}{4} I_{n-1}$$

$$\int_1^e x^3 (\ln x)^2 dx = I_2$$

$$I_2 = \frac{e^4}{4} - \frac{2}{4} I_1$$

$$I_1 = \frac{e^4}{4} - \frac{1}{4} I_0$$

$$I_0 = \int_1^e x^3 dx$$

$$= \left[\frac{x^4}{4} \right]_1^e$$

$$= \left(\frac{e^4 - 1}{4} \right)$$

$$I_1 = \frac{e^4}{4} - \frac{(e^4 - 1)}{16}$$

$$I_2 = \frac{e^4}{4} - \frac{1}{2} \left[\frac{e^4}{4} - \frac{(e^4 - 1)}{16} \right]$$

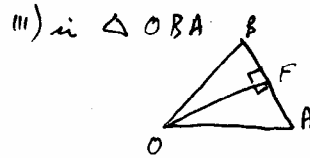
$$= \frac{e^4}{4} - \frac{e^4}{8} + \frac{e^4}{32} - \frac{1}{32}$$

$$= \frac{5e^4 - 1}{32}$$

$$8b) i) z_2 = 2 \angle 4\frac{\pi}{5}$$

$$ii) (z_2)^5 = 2^5 \angle 20\frac{\pi}{5}$$

$$= 32.$$



Centroid $OF \perp BA$

$$\angle FOA = \frac{1}{2} \cdot 20\frac{\pi}{5} = \frac{\pi}{5}$$

$$\sin \frac{\pi}{5} = \frac{FA}{OA} = \frac{FA}{2}$$

$$\therefore FA = 2 \sin \frac{\pi}{5}$$

$$BA = 2FA = 4 \sin \frac{\pi}{5}$$

$$\therefore \text{perimeter of polygon} = 5 \times 4 \sin \frac{\pi}{5}$$

$$= 20 \sin \frac{\pi}{5}$$

c By De Moivre's Th

$$\begin{aligned} (\cos 5\theta + i \sin 5\theta) &= (\cos \theta + i \sin \theta)^5 \\ &= (C + iS)^5 \\ &= C^5 + 5iC^4S + 10i^2C^3S^2 + 10i^3C^2S^3 + 5i^4CS^4 + i^5S^5 \\ &= (C^5 - 10C^3S^2 + 5CS^4) + i(5C^4S - 10C^2S^3 + S^5) \end{aligned}$$

$$\therefore \cos 5\theta = \cos^5 \theta - 10\cos^3 \theta \sin^2 \theta + 5\cos \theta \sin^4 \theta$$

$$\sin 5\theta = 5\cos^4 \theta \sin \theta - 10\cos^2 \theta \sin^3 \theta + \sin^5 \theta$$

$$\therefore \tan 5\theta = \frac{5\cos^4 \theta \sin \theta - 10\cos^2 \theta \sin^3 \theta + \sin^5 \theta}{\cos^5 \theta - 10\cos^3 \theta \sin^2 \theta + 5\cos \theta \sin^4 \theta}$$

÷ +ve by $\cos^5 \theta$

$$\therefore \tan 5\theta = \frac{5\tan \theta - 10\tan^3 \theta + \tan^5 \theta}{1 - 10\tan^2 \theta + 5\tan^4 \theta}$$

$$x^5 - 5x^4 - 10x^3 + 10x^2 + 5x - 1 = 0$$

$$x^5 - 10x^3 + 5x = 1 - 10x^2 + 5x^4$$

$$\therefore \frac{x^5 - 10x^3 + 5x}{1 - 10x^2 + 5x^4} = 1$$

let $x = \tan \theta \therefore \tan 5\theta = 1$

$$5\theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \frac{13\pi}{4}, \frac{17\pi}{4}$$

$$\theta = \frac{\pi}{20}, \frac{5\pi}{20}, \frac{9\pi}{20}, \frac{13\pi}{20}, \frac{17\pi}{20}$$

$$\therefore x = \tan \frac{\pi}{20}, \tan \frac{\pi}{4}, \tan \frac{9\pi}{20}, \tan \frac{13\pi}{20}, \tan \frac{17\pi}{20}$$

$$\text{now } \tan \frac{\pi}{4} = 1 \quad \tan \frac{13\pi}{20} = \tan \frac{33\pi}{20}$$

$$\therefore x = \tan \frac{\pi}{20}, \tan \frac{9\pi}{20}, \tan \frac{17\pi}{20}, \tan \frac{33\pi}{20}, 1$$

from polynomial $5x^5 - 5 = 0$

$$\therefore \tan \frac{\pi}{20} + \tan \frac{9\pi}{20} + \tan \frac{17\pi}{20} + \tan \frac{33\pi}{20} + 1 = 5$$

$$\therefore (\quad) = 4$$